Bhuvesh **Kumar**

Senior Research Scientist · Ph.D. in Computer Science

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Experience

Snap Inc. *Bellevue, WA*
Senior Research Scientist *Sept. 2024 - Present*

- Develop large language and multimodal models for generative recommendation, agentic search and retrieval, content understanding, and cross-domain user modeling. Created and serve as technical lead for Snap's GenAI recommendation library, used by all production generative recommendation and multimodal Semantic ID models at Snap and by 30+ scientists and ML engineers.
- Developed multimodal Semantic IDs that incorporate collaborative signals. These representations are used as ID features in 20+ production models across Content, Ads, Lens, Search, and product recommendation, and as targets for all production generative recommendation models at Snap.
- Conducted research on post-training strategies and led implementation of LLM-based generative recommendation with Semantic IDs, contributing to multiple production launches that improved total content views and daily active users.
- Designed the inference pipeline for Semantic ID beam-search decoding using a custom CuTe kernel, enabling 4x more beams and saving over $2.4 million annually in generative recommendation inference costs.
- Extended the GenAI library for distributed training of large-scale sequential recommenders with sharded embedding tables using TorchRec and NVIDIA DynamicEmb. Migrated Snap's Universal User Embedding Model from TensorFlow to this stack, saving at least $1.6 million annually; its embeddings serve as foundational user representations in 24+ production models.
- Co-created GRID, the open-source subset of the platform; its accompanying paper received the CIKM 2025 Best Paper Award, and the project has earned 700+ GitHub stars.

TikTok *Bellevue, WA*
Machine Learning Scientist *Jan. 2023 - Sept. 2024*

- Developed large-scale sequential user models and personalization algorithms for TikTok U.S. Core Recommendation.
- Introduced and owned a new final re-ranking stage that moved TikTok U.S. recommendation from point-wise item scoring to list/session-level Transformer sequence modeling, increasing total watch time by 2.4%.
- Designed a personalized mixing stage for organic, cold-start, and commercial content; coordinated with horizontal teams to preserve cold-start and commercial-content distribution goals while optimizing long-term user retention.
- Led the migration of TikTok U.S. Friends, Hashtag, and Social recommendation services from a single data center to a disaster-tolerant multi-data-center design.

Amazon *Seattle, WA*
Applied Scientist Intern *May 2021 - Aug. 2021*

- Introduced a group-fairness notion for multi-armed bandits and developed a no-regret algorithm balancing fairness and performance.

University of Washington *Seattle, WA*
Visiting Research Scientist *Jan. 2021 - Apr. 2021*

- Designed incentive-compatible, differentially private auction mechanisms for revenue maximization in repeated auctions.

Meta *Menlo Park, CA*
Research Intern, Core Data Science *May 2020 - Nov. 2020*

- Developed optimal-spend-plan estimation and automatic bid-pacing algorithms for budget-constrained advertising campaigns.

Meta *Menlo Park, CA*
Machine Learning Engineering Intern *May 2019 - July 2019*

- Improved Dynamic Product Ads ranking and increased ad revenue while reducing model memory by 60 TB (86%).

Georgia Institute of Technology *Atlanta, GA*
Graduate Research Assistant *Aug. 2017 - Dec. 2022*

- Developed robust learning algorithms with provable guarantees using online learning, algorithmic game theory, and differential privacy.

Education

Georgia Institute of Technology *Atlanta, Georgia*
Ph.D. in Computer Science (Advisors: Dr. Jacob Abernethy and Dr. Jamie Morgenstern) *2017 - 2022*

Georgia Institute of Technology *Atlanta, Georgia*
M.S. in Computer Science (Specialization in Machine Learning, GPA: 4.0/4.0) *2017 - 2022*

Indian Institute of Technology, Kanpur *Kanpur, India*
B.Tech in Computer Science and Engineering (GPA: 9.7/10) *2013 - 2017*

Selected Recent Publications

- **[13] Semantic IDs for Recommender Systems at Snapchat: Use Cases, Technical Challenges, and Design Choices**
  SIGIR 2026
  C. M. Ju, T. Zhao, L. Neves, L. Collins, B. Kumar, et al.
- **[12] LLM-Based Generative Retrieval for Snapchat Content Recommendation**
  2026 Preprint
  L. Collins, J. Ren, D. Loveland, B. Kumar, et al.
- **[11] Exploiting ID-Text Complementarity via Ensembling for Sequential Recommendation**
  RecSys 2026
  L. Collins, B. Kumar, C. M. Ju, T. Zhao, D. Loveland, L. Neves, N. Shah
- **[10] Training-Free LLM-Based Recommendation with Post-LLM Item Refinement Using Collaborative Signals**
  CIKM 2026
  K. Kim, S. Kim, G. Lee, S. Kang, S. Kim, L. Collins, B. Kumar, et al.
- **[9] Sequential Data Augmentation for Generative Recommendation**
  WSDM 2026
  G. Lee, B. Kumar, C. M. Ju, T. Zhao, K. Shin, N. Shah, L. Collins
- **[8] CoSearch: Joint Training of Reasoning and Document Ranking via Reinforcement Learning for Agentic Search**
  COLM 2026
  H. Zeng, L. Collins, B. Kumar, N. Shah, H. Zamani
- **[7] Learning Universal User Representations Leveraging Cross-Domain User Intent at Snapchat**
  SIGIR 2025
  C. M. Ju, L. Neves, B. Kumar, L. Collins, T. Zhao, et al.
- **[6] Generative Recommendation with Semantic IDs: A Practitioner's Handbook**
  CIKM 2025
  C. M. Ju, L. Collins, L. Neves, B. Kumar, L. Y. Wang, T. Zhao, N. Shah
  Best Paper Award; co-creator of GRID (700+ stars on GitHub)
- **[5] Revisiting self-attention for cross-domain sequential recommendation**
  KDD 2025
  C. M. Ju, L. Neves, B. Kumar, L. Collins, T. Zhao, et al.
- **[4] Private Mechanism Design via Quantile Estimation**
  ICLR 2025
  Y. Yang, T. Xiao, B. Kumar, J. Morgenstern
- **[3] ActiveHedge: Hedge meets Active Learning**
  ICML 2022 (Spotlight)
  B. Kumar, J. Abernethy, V. Saligrama
- **[2] Observation Free Attacks on Stochastic Bandits**
  NeurIPS 2021
  Y. Xu, B. Kumar, J. Abernethy
- **[1] Learning Auctions with Robust Incentive Guarantees**
  NeurIPS 2019
  B. Kumar, J. Abernethy, R. Cummings, J. Morgenstern, S. Taggart

Complete publication list: Google Scholar.

Technical Skills

Programming & ML	Python, C++, C; PyTorch, TorchRec, TensorFlow, verl
Distributed ML & Data	Ray, vLLM, TensorRT-LLM; GCP (GKE, Vertex AI); SQL, Spark, Flink, Hive

Awards

2025	**Best Paper Award**, ACM CIKM
2019	**NeurIPS Travel Award**, Neural Information Processing Systems
2017	**Chair's Fellowship**, Georgia Institute of Technology
2014-16	**Academic Excellence Award (Dean's List)**, IIT Kanpur
2011	**KVPY Fellowship**, Government of India
2009	**NTSE Scholarship**, Government of India

Academic Service

Reviewer / Program Committee: ACL (2026); ICLR (2025, 2026); NeurIPS (2020, 2021, 2025); COLT (2020, 2021, 2023); SODA (2021, 2023); ICML (2022); JMLR (2021); ALT (2018, 2020).